Single-Spectrum-Driven Binary-Class Sparse Representation Target Detector for Hyperspectral Imagery

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***Abstract*— In this article, a single-spectrum-driven binary-class sparse representation target detector (SSBSTD) via target and background dictionary construction (BDC) is proposed. The SSBSTD leans upon the binary-class sparse representation (BSR) model. Due to the fact that a background spectrum usually consists in background samples composed low-dimensional subspace and a target spectrum also consists in target samples composed low-dimensional subspace, only background samples should be used for sparsely representing the test pixel under the target absent hypothesis and the samples from target-only dictionary for target present hypothesis. To alleviate the problem that there are insufficient available target samples in the sparse representation model, this article proposed a predetection method to construct the target dictionary utilizing the given target spectrum. With regard to the BDC, we proposed an approach based on the classification to generate a global over-complete background dictionary. The detection output is composed of the residual difference between the BSR. Extensive experiments were made on four benchmark hyperspectral images and the experimental results indicate that our SSBSTD algorithm demonstrates superior detection performances.**

***Index Terms*— Binary hypothesis, hyperspectral imagery (HSI), sparse representation, target detection.**

I. Introduction

TARGET detection for hyperspectral imagery (HSI) is basically a binary classifier. It attempts to classify the pixels in HSI as either target or background according to the spectral characteristics [1]–[6]. Target detection has been successful in many fields, such as civil and military, and it has developed for numerous applications, such as surveillance and reconnaissance [7]–[11]. However, the traditional classification algorithms are inapplicable in target detection for the reason that the objects of interest account for only a small part and background land covers account for a large part. Therefore, the Neyman–Pearson criterion is widely used in traditional target detection, as it is capable of maximizing the detectivity under a fixed false alarm rate [12].

Over the past several decades, numerous classical target detectors have been developed and applied to hyperspectral images [13]–[18]. In these classical target detectors, spectral angle mapper (SAM) [19] is probably the simplest one. SAM achieves the output by evaluating the spectral angle between the test pixels and the given target pixel. Based on the linear mixing model, adaptive coherence estimator (ACE) [20], [21], orthogonal subspace projection (OSP) [22]–[25], and matched subspace detector (MSD) [26] have been successfully applied. Adaptive matched filter (AMF) [27], [28] and spectral matched filter (SMF) [29] are well-known detectors utilizing spectral signature for matching. Another classical detector, constrained energy minimization (CEM) [23], [30], achieves the output result by filtering techniques, which is an effective method to highlight the targets in HSI. In these algorithms, the estimation of background covariance is the most important factor for the detection performance [31]. However, these algorithms assumed noise as a normal distribution with a zero mean when estimating the background [32]. As a matter of fact, the noise in HSI may not be in the case. Therefore, the detection performances of the classical target detectors are usually unsatisfactory.

Over the years, many sparse representation-based target detectors have been developed in HSI. In these detectors, few samples in an over-complete dictionary composed of training samples is capable of approximately representing the test sample [8]. Following the concept, the basic sparsity-based target detector (STD) [33] is presented, in which the samples lie in the union dictionary are utilized to express the test sample. The obtained background sparse vector and the target sparse vector are indirectly used for the detection decision at the same time. The sparse vectors are recovered using orthogonal matching pursuit (OMP) sparse-coding method [34], [35]. To utilize both spatial and spectral information in HSI, the simultaneous sparsity-based target detector (SSTD) [36] is proposed. SSTD incorporates the joint sparse model to fuse the spatial information of adjacent pixels. However, when SSTD detects the central pixel, it is assumed that the contribution

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of all adjacent pixels is absolutely equal. It is unsuitable in some heterogeneous areas. By comparing the similarity between adjacent pixels and the central pixel, the spatially adaptive sparsity-based target detector (SASTD) [37] alleviates this problem. Another target detector proposed via sparsity characteristics in HSI is the hybrid sparsity and distance-based algorithm (HSDD) [31]. By building a pure background dictionary utilizing discriminative dictionary learning, it obtains the final output on the basis of the STD detector. Given the dominance of the binary hypothesis, the well-known sparsity representation-based binary hypothesis (SRBBH) detector [38] is proposed, in which only background training samples are used for representing background pixels when targets are absent and union training samples are utilized for approximating target pixels when target present. The sparse vectors under two different hypotheses are used for the final detection decision. Following the SRBBH, the hybrid sparsity and statistics-based detector (HSSD) [39] that hybrid sparsity and statistics characteristics is proposed. Different from SRBBH, HSSD assumed that the pixels in HSI are in a Gaussian distribution and they perform the same structure in covariance when target absent, but they have a difference in variances when target present.

Although numbers of sparsity-based algorithms have been successfully applied for target detection, there are still some drawbacks in these models. The most important one is that they all use the union dictionary to express both background and target pixels, where the two sparse vectors are simultaneously recovered. In the processing of recovering the sparse vector, the two sparse vectors will affect each other and result in weakened detection performance. Even in binary hypothesis-based sparse representation detectors such as SRBBH and HSSD, they try to model the two-class pixels, respectively, but there is also a union dictionary under the alternative hypothesis. Since a target spectrum will not lie in background samples composed low-dimensional subspace, using numbers of background training samples to represent a target pixel is unreasonable, which also causes a poor detection performance. In addition, the present-day sparsity-based detectors usually use the pixels selected from the HSI as the target prior to constructing the target dictionary. It is an unreasonable way since we do not know which pixel in the HSI is the target. Besides, the pixels selected in this way are not enough. As for background dictionary construction (BDC), sparse representation-based methods tend to construct it via a dual concentric window. Since there are a variety of features in the HSI, the local background dictionary may be incomplete.

To overcome these shortcomings, we proposed a single spectrum-driven binary-class sparse representation target detector (SSBSTD). In SSBSTD, we use the samples in the background dictionary to sparsely represent a background sample under the target absent hypothesis and the samples in the target dictionary are utilized to sparsely represent a target sample under the target present hypothesis. To solve the problem that there are usually insufficient target samples in the sparsity-based model, we proposed a predetection method to construct the target dictionary utilizing the given target prior. As for BDC, we proposed a new approach based on the classification to generate a global over-complete background dictionary. Taking advantage of the nonlocal similarity among the local regions, it can guarantee that background pixels in HSI could obtain a fairly good representation via the global background dictionary. The detection output of SSBSTD is composed of the residual difference between the binary-class sparse representation (BSR). The following are the contributions of this article.

1) A BSR-based model is proposed denoted as BSR in this article. The BSR only uses background training samples to represent background pixels and target training samples to target pixels. It avoids the problem that target and background sparse vectors affect each other in the processing of recovering sparse vector, which is able to help separate target pixels and background pixels.
2) By utilizing a classical detector for predetection, our method only needs a given target prior to detecting targets, while the other sparse representation-based detectors need target pixels selected in the HSI as input prior information. Since we do not know which pixel in the HSI is the target, it is an unreasonable way to choose target pixels from the HSI as the input prior information. Besides, our method can obtain enough target training samples via predetection, which contributes to a good representation of target pixels, and finally enhances the detection efficiency.
3) An approach based on the classification to generate a global over-complete background dictionary is proposed. The global over-complete background dictionary is more complete when compared with the local background dictionary, which can guarantee each background pixel in the image scene obtains a better representation. Hence, our SSBSTD can obtain better detection performance.

The rest of this article is organized as follows. Section II first provides a detailed description of the SSBSTD. Extensive experiments are provided to test the efficiency of the SSBSTD in Section III. Finally, concluding remarks are provided in Section IV.

II. Single-Spectrum-Driven BSR Target Detector

A. Binary-Class Sparse Representation

Inspired by the basic sparse representation-based model that a sample of HSI can be approximated via very few elements in the over-complete dictionary, a BSR model is proposed. Consider the test sample $\mathbf{x}$ to be a B-dimensional vector expressing various spectral bands. If the test sample is a background pixel, its spectrum will lie in the training samples composed subspace from the background dictionary, and the test sample $\mathbf{x}$ can then be approximated as follows:

$$\begin{aligned}\mathbf{x} &\approx \alpha_1\mathbf{a}_1^b + \alpha_2\mathbf{a}_2^b + \cdots + \alpha_{N_b}\mathbf{a}_{N_b}^b \\ &= \left[\mathbf{a}_1^b\ \mathbf{a}_2^b\ \ldots\ \mathbf{a}_{N_b}^b\right]\left[\alpha_1\ \alpha_2\ \ldots\ \alpha_{N_b}\right]^T \\ &= \mathbf{A}^b\boldsymbol{\alpha}\end{aligned} \tag{1}$$

where $\boldsymbol{\alpha}$ shows the abundances of the samples in the $\mathbf{A}^b$, $\mathbf{A}^b$ refers to the background dictionary, and N_b is the number of samples in $\mathbf{A}^b$. Uniformly, if $\mathbf{x}$ is a target pixel, its spectrum

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could approximately lie in the target samples composed subspace from the target-only dictionary and can be written as

$$\begin{aligned}\mathbf{x} &\approx \beta_1 \mathbf{a}_1^t + \beta_2 \mathbf{a}_2^t + \cdots + \beta_{N_t} \mathbf{a}_{N_t}^t \\ &= \left[\, \mathbf{a}_1^t \ \mathbf{a}_2^t \ \ldots \ \mathbf{a}_{N_t}^t \,\right] \left[\, \beta_1 \ \beta_2 \ \ldots \ \beta_{N_t} \,\right]^T \\ &= \mathbf{A}^t \boldsymbol{\beta} \end{aligned} \tag{2}$$

where $\boldsymbol{\beta}$ expresses the abundances of the training samples in the $\mathbf{A}^t$, $\mathbf{A}^t$ refers to the target dictionary, and N_t is the number of samples in $\mathbf{A}^t$.

In summary, an undefined sample is deemed to lie in either target or background subspace in BSR. Therefore, an undefined sample is selected from binary hypotheses can be written as

$$\begin{aligned} H_0 &: \mathbf{x} = \mathbf{A}^b \boldsymbol{\alpha} + \boldsymbol{\varepsilon}_0, \quad \text{target absent} \\ H_1 &: \mathbf{x} = \mathbf{A}^t \boldsymbol{\beta} + \boldsymbol{\varepsilon}_1, \quad \text{target present} \end{aligned} \tag{3}$$

where $\boldsymbol{\varepsilon}_0$ and $\boldsymbol{\varepsilon}_1$ are the approximation residuals.

In other binary hypothesis-based sparse representation models like SRBBH, sparsity constraint employed in the two hypotheses is usually set as the same. In our proposed detector, since we use a global over-complete background dictionary $\mathbf{A}^b$ to sparsely represent background pixels instead of a local one, a higher sparsity level should be used to make sure an adequate fitting for each background pixel. Therefore, the sparsity constraint Kb is employed for H_0 and Kt for H_1. The sparsity constraint of our proposed detector will be further discussed and examined in Section III.

In BSR, the important work is to reconstruct the sparse vector of samples, it can be described as follows. Given the background dictionary, we can get the background sparse vector $\boldsymbol{\alpha}$ by settling the following equation:

$$\widehat{\boldsymbol{\alpha}} = \arg\min \ \|\mathbf{A}^b \boldsymbol{\alpha} - \mathbf{x}\|_2 \quad \text{subject to } \|\boldsymbol{\alpha}\|_0 \leq Kb \tag{4}$$

where $\widehat{\boldsymbol{\alpha}}$ is the recovered background sparse vector, $||\cdot||_2$ denotes the ℓ_2-norm, and $||\cdot||_0$ denotes the ℓ_0-norm. In this article, ℓ_0-norm expresses the number of nonzero entries in $\widehat{\boldsymbol{\alpha}}$ [33] and we define Kb as the given upper limit of the sparse level [40]. Solving (4) is basically a recovery process, we use the OMP [34], [35], [41] method to obtain the sparse vector. Similarly, we can get the target sparse vector $\boldsymbol{\beta}$ by settling the following equation:

$$\widehat{\boldsymbol{\beta}} = \arg\min \ \|\mathbf{A}^t \boldsymbol{\beta} - \mathbf{x}\|_2 \quad \text{subject to } \|\boldsymbol{\beta}\|_0 \leq Kt \tag{5}$$

where $\widehat{\boldsymbol{\beta}}$ is the recovered target sparse vector. Similar to (4), we also use the OMP to settle (5) and get the sparse vector. Once $\widehat{\boldsymbol{\alpha}}$ and $\widehat{\boldsymbol{\beta}}$ are obtained, then we calculate the residuals of recovery by

$$\begin{aligned} r_0(x) &= \|\mathbf{x} - \mathbf{A}^b \widehat{\boldsymbol{\alpha}}\|_2 \\ r_1(x) &= \|\mathbf{x} - \mathbf{A}^t \widehat{\boldsymbol{\beta}}\|_2. \end{aligned} \tag{6}$$

Finally, the output of the BSR model can be obtained by

$$D_{\mathrm{BSR}}(x) = r_0(x) - r_1(x). \tag{7}$$

Noticed the formulation in (7), the detection output is absolutely based on the approximate residuals. However, there are two termination conditions in the greedy algorithm OMP: The preset level of sparsity is reached or sufficiently small residual is reached. In our proposed BSR model, we use the first one as the stopping criterion. Because the detection output is completely based on the residuals in two hypotheses. If the stopping criterion becomes that sufficiently small residual is reached, r_0 and r_1 are the values that we set in advance. Due to this case, the second criterion should be removed. Therefore, the sparsity level becomes a critical parameter which controls the approximation accuracy.

Compared with the STD model and the SRBBH model, our proposed BSR model has a better discriminative ability to detect the targets. In STD, the union dictionary includes target and background samples. When the test pixel is represented, samples in the union dictionary are all used to express the test pixel. Therefore, the target and background sparse vectors are simultaneously recovered. During the process of recovering, the two sparse vectors will interact with each other, which is easy to cause a nonzero sparse vector for target and also affect the background sparse vector. Then it will cause a poor representation for both target and background pixels, and finally obtain a bad detection performance. In SRBBH, it tries to model the test sample separately. However, there still contains both background and target samples under the target present hypothesis. Since a target spectrum will not consist in the background samples composed subspace, the background samples cannot be used to represent a target pixel. Therefore, using numbers of background samples to represent the target pixel is unreasonable, which causes a poor representation for target pixels. As a result, SRBBH also obtains an unsatisfactory detection performance.

In our proposed BSR, we consider the target and background pixels separately. The samples that only exist in the background dictionary are utilized to sparsely express the test sample under the target absent hypothesis. The samples that only occur in the target dictionary are used to sparsely express the test sample under the target present hypothesis. To be specific, we only use the target dictionary to sparsely represent target pixels and background dictionary to background pixels. The two recovery processes of target and background sparse vectors are independent of each other, and a better representation of two classes will be provided. Therefore, BSR has enough discriminative ability to detect the targets. In the BSR model, target dictionary and background dictionary are used to represent target and background pixels, respectively, which are two crucial portions for BSR. Thus, we provide a detailed description of target and BDC in the following part.

B. Target Dictionary Construction (TDC)

According to the aforesaid depiction about the BSR model, TDC is important for the SSBSTD detector. However, there are insufficient target training samples in the sparse representation-based method. The present-day sparse representation-based target detection methods usually use the pixels selected from the HSI as the target prior to the TDC. In hyperspectral target detection tasks, we usually do not

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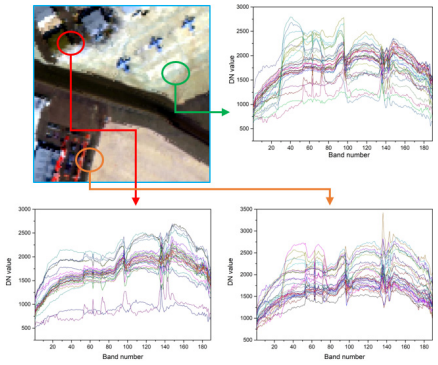

Fig. 1. Spectrum of pixels in the local background regions with the AVIRIS data set.

know which pixel in the image is the target pixel, so the way that selecting target pixels in HSI is improper. Apart from this, we cannot obtain enough target pixels for dictionary construction in such a way. Therefore, there are not enough samples for expressing each target pixel in the image scene. Herein, a novel TDC method is presented using only a given target spectrum.

In the beginning, we utilize a simple and classical detector for predetection by the given target spectrum. In this article, the CEM detector is selected for predetection and other classical detectors, that is ACE, SAM, and SMF can also be used for predetection. Then the detection output of CEM is achieved and the output value of each pixel in HSI can be obtained. Consider a hyperspectral image with n pixels $\boldsymbol{\Omega} = \{\boldsymbol{\omega}_1, \boldsymbol{\omega}_2, \ldots, \boldsymbol{\omega}_n\}$, we can get n output values. The pixel with larger output value is more likely to be a target sample; we choose these pixels corresponding larger output values of CEM to form the target dictionary. Specifically, these output values are first sorted in descending order, and the pixels corresponding output values become $\boldsymbol{\Omega} = \{\boldsymbol{\varphi}_1, \boldsymbol{\varphi}_2, \ldots, \boldsymbol{\varphi}_n\}$ where the output value of the pixel $\boldsymbol{\varphi}_1 \geq \boldsymbol{\varphi}_2 \geq \cdots \geq \boldsymbol{\varphi}_n$; we select the pixels $\boldsymbol{\Phi} = \{\boldsymbol{\varphi}_1, \boldsymbol{\varphi}_2, \ldots, \boldsymbol{\varphi}_k\}$ for TDC. Herein, $k = t * n_t$, where n_t is the real target pixels' number of the HSI and t controls the target samples' number in the target dictionary. In this way, adequate target training samples can be obtained for constructing the target dictionary, each target pixel in the image scene gets a good representation, and finally, the detection efficiency can be enhanced. As a result, the parameter t becomes a critical determinant to control the number of target samples. For different HSIs, there exists a different optimal t. In this article, the optimum t is obtained for different HSIs based on previous experiences [42].

C. Background Dictionary Construction

In SSBSTD, constructing an over-complete background dictionary is also a challenge. In [43], dual concentric window is utilized for modeling the local background and used for forming a background dictionary. Inspired by the dual concentric window, we analyze the local background; Fig. 1 shows the spectrum of pixels in the local background regions. We find that the spectrums of pixels vary greatly from one to another even in the same local background region, which indicates that they belong to different classes. As a result, in some local areas, when the class of the center pixel is different from most others in the local area, this center pixel will get a bad representation, which finally causes a deteriorated detection performance. On the other hand, some pixels show the same spectrum even in different local regions. There exists a nonlocal similarity between the local regions [44], thus we want to take advantage of it when constructing the background dictionary. Hence, a novel method based on the classification to generate a global over-complete background dictionary is proposed.

First of all, for a given hyperspectral image data, we first apply the principal component analysis (PCA) on the data set. Then the pixels of HSI are grouped into p clusters via the k-means clustering algorithm. The initial seeds of the k-means clustering are random and the number of clusters is set as 10 in this article. For each cluster, there are p_i pixels which can be expressed by

$$\mathbf{E}_i = \{\mathbf{e}_{i1}, \mathbf{e}_{i2}, \ldots, \mathbf{e}_{ip_i}\}, \quad 1 \leq i \leq p. \tag{8}$$

For the ith cluster, each pixel $\mathbf{e}_{\text{is}}$ where $1 \leq s \leq p_i$ can be approximated by the other pixels $\mathbf{E}'_i$ in this cluster which can be written as

$$\mathbf{e}_{\text{is}} = \mathbf{E}'_i \boldsymbol{\theta}_{\text{is}}, \quad 1 \leq i \leq p, \ 1 \leq s \leq p_i \tag{9}$$

$$\mathbf{E}'_i = \{\mathbf{e}_{i1}, \mathbf{e}_{i2}, \ldots, \mathbf{e}_{i(s-1)}, \mathbf{e}_{i(s+1)}, \ldots, \mathbf{e}_{ip_i}\}, \quad 1 \leq i \leq p \tag{10}$$

where $\boldsymbol{\theta}_{\text{is}}$ is a sparse vector and it indicates the frequency that $\mathbf{E}'_i$ occurs in sparsely representing $\mathbf{e}_{\text{is}}$ [45]. The corresponding $\mathbf{e}_{\text{is}}$ is a more typical pixel in the ith cluster if $\boldsymbol{\theta}_{\text{is}}$ has a higher frequency [46]. Therefore, we choose these typical pixels $\boldsymbol{\delta}_i = \{\boldsymbol{\delta}_{i1}, \boldsymbol{\delta}_{i2}, \ldots, \boldsymbol{\delta}_{ij_i}\}$ in the ith cluster as background training samples$\boldsymbol{\delta} = \{\boldsymbol{\delta}_1, \boldsymbol{\delta}_2, \ldots, \boldsymbol{\delta}_p\}$ to construct a global over-complete background dictionary. The total number of background training samples N_b selected from each cluster can be written as

$$N_b = j_1 + j_2 + \cdots + j_p. \tag{11}$$

In (11), j_i means the number of selected samples in the ith cluster and it can be calculated by

$$j_i = \frac{n_i}{n} * N_b \tag{12}$$

where n is the total number of pixels in the image and n_i is the total number of pixels in the ith cluster.

Using a global over-complete background dictionary formed via the aforementioned method takes advantage of the nonlocal similarity between the local regions. Moreover, for the reason that the objects of interest account for only a small part and background land covers account for a large part, the target pixel cannot be a typical pixel when building a global over-complete background dictionary. Therefore, the background dictionary constructed by this method is not only complete but also pure, it leads to better representation for each background pixel in HSI, and finally achieves a good detection performance. As a result, the amount of background samples N_b selected in each cluster is a critical parameter to form the background dictionary. Some parametric analysis will be made to seek out the optimum N_b of HSI in the Section III.

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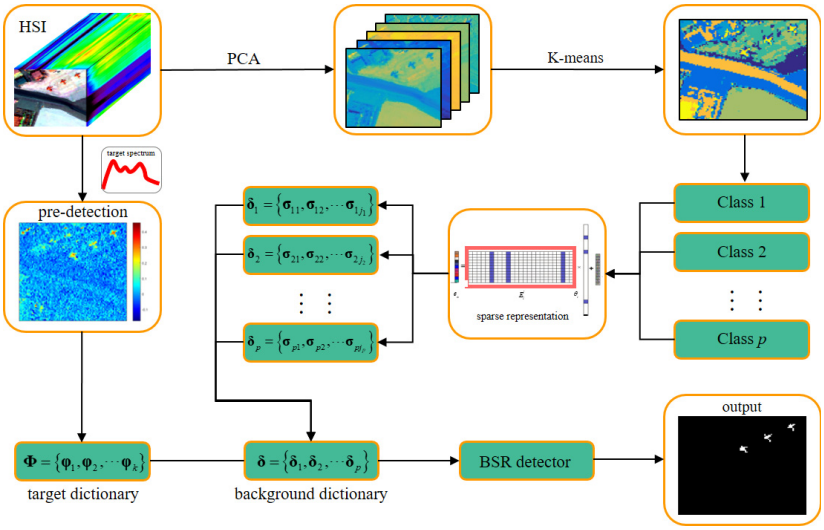


Fig. 2. Implementation flowchart of SSBSTD.

D. Final Scheme of Proposed Target Detector

On the basis of the aforesaid depiction, the implementation steps of the SSBSTD are introduced in the following and Fig. 2 shows the implementation flowchart of the SSBSTD.

Step 1: Input the HSI with n pixels as samples and a given target spectrum as prior information.

Step 2: The classical detector CEM is used for predetection based on the given target spectrum and then sort the pixels $\mathbf{\Omega} = \{\boldsymbol{\varphi}_1, \boldsymbol{\varphi}_2, \ldots, \boldsymbol{\varphi}_n\}$ with output values in descending order. The pixels $\mathbf{\Phi} = \{\boldsymbol{\varphi}_1, \boldsymbol{\varphi}_2, \ldots, \boldsymbol{\varphi}_k\}$ are utilized for forming the target dictionary $\mathbf{A}^t$.

Step 3: A feature reduction algorithm PCA and the k-means clustering approach are used in the HSI, then the pixels of HSI are grouped into p clusters.

Step 4: For the ith cluster, each pixel $\mathbf{e}_{\text{is}}$ is sparsely represented by other pixels $\mathbf{E}'_i$ in this cluster and the sparse vector $\boldsymbol{\theta}_{\text{is}}$ is obtained.

Step 5: Calculate the frequency of $\boldsymbol{\theta}_{\text{is}}$ and choose the typical pixels $\boldsymbol{\delta} = \{\boldsymbol{\delta}_1, \boldsymbol{\delta}_2, \ldots, \boldsymbol{\delta}_p\}$ in each cluster to form the background dictionary $\mathbf{A}^b$.

Step 6: Solve the optimization problem in (4) and (5) using $\mathbf{A}^b$ and $\mathbf{A}^t$, then obtain the sparse vectors $\widehat{\boldsymbol{\beta}}$ and $\widehat{\boldsymbol{\alpha}}$.

Step 7: Calculate the residuals of recovery under two hypotheses, the output values of detection can be obtained via BSR model in (7).

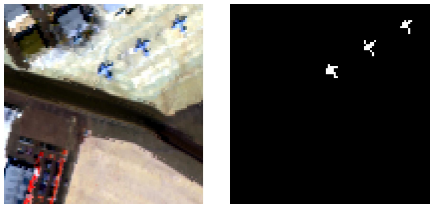
Step 8: Output the detection results of the SSBSTD.

III. Experimental Results and Analysis

In this part, numerous experiments on four benchmark HSIs are made to assess the validity of the SSBSTD algorithm. Four benchmark employed HSIs are first introduced in detail, and then parameter analyses on the sparse level and the number of background samples for the SSBSTD algorithm are provided. Finally, we present the detection performance and analyses.

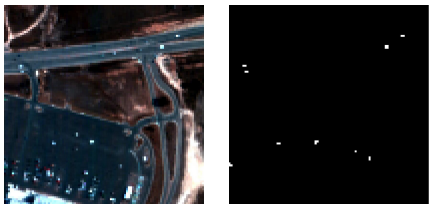
A. Hyperspectral Data Set

The first benchmark data set is an airborne visible/infrared imaging spectrometer (AVIRIS) assembled HSI from San Diego. There are 224 spectral bands in the image and the resolution of this image is 3.5 m. In the AVIRIS data set, the wavelength range is 370–2510 nm. Due to the problem of the water absorption band and the low SNR band, 35 bad bands are discarded and 189 bands are retained in the data [1], [42]. In this article, a suburban scene that covers 100 $*$ 100 pixels is used for the experiment, as shown in Fig. 3(a). There are 58 pixels distributed in three planes and they are regarded as targets needed to detect which are illustrated in Fig. 4(b).



(a) (b)

Fig. 3. AVIRIS hyperspectral data. (a) Image scene. (b) Ground-truth.



(a) (b)

Fig. 4. HYDICE hyperspectral data. (a) Image scene. (b) Ground-truth.

The second HSI is assembled by the hyperspectral digital imagery collection experiment (HYDICE) sensor. The spatial resolution and spectral resolution of this data is 2 m and 10 nm, respectively. The HSI has 210 spectral bands and its wavelengths range is 400–2500 nm. About 48 bad bands are discarded and 162 bands are retained for the detection experiments [47]. From this data, a suburban scene that covers 100 $*$ 100 pixels is selected to evaluate the effectiveness of

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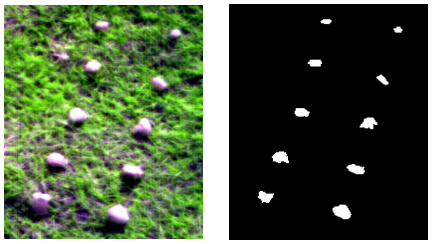

Fig. 5. Cri hyperspectral data. (a) Image scene. (b) Ground-truth.

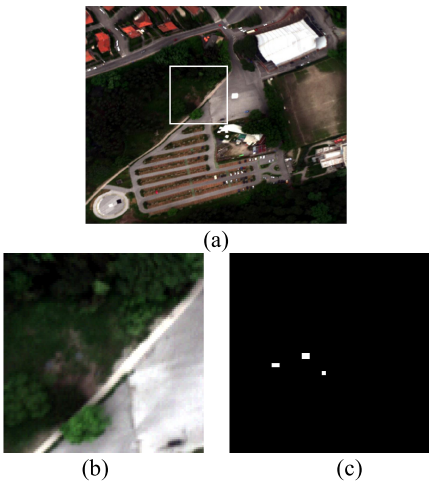

Fig. 6. Viareggio hyperspectral data. (a) Image scene. (b) Subimage scene. (c) Ground-truth of subimage scene.

the SSBSTD algorithm, and there are several vehicles that have 21 pixels selected as the targets to be detected. Fig. 4 shows the HYDICE data.

The third data set used in the experiment is assembled via the Nuance Cri hyperspectral sensor and its spectral resolution is 10 nm [48]. The wavelengths range of this data set is 650–1100 nm and there are 46 spectral bands. The HSI covers an area of 280 $*$ 200 pixels as shown in Fig. 5(a). Herein, the 10 rocks containing anomalous pixels different from the grassy background are illustrated in Fig. 5(b). About 1254 pixels are regarded as targets to be detected in this data set [49].

The last benchmark data are collected in a HSI data collection campaign that called the "Viareggio 2013 trial" [50]. Data sets in the campaign are assembled via a push-broom sensor called the "Sistema Iperspettrale Modulare Galileo Avionica" [51]. In the obtained data sets, we select the D1F12H1 as experimental data. The subset of D1F12HI has 375 $*$ 450 pixels and covers a parking lot, a football field, and several sports facility buildings, as shown in Fig. 6(a). The wavelengths range of this data set are 400–1000 nm and there are 511 spectral bands. The spatial resolution and spectral resolution of this data are 0.6 m and 2.3 nm, respectively. Herein, a subscene that covers 100 $*$ 100 pixels is opted to assess the performance of the SSBSTD as shown in Fig. 6(b). There are three panels containing 24 pixels regarded as targets which are needed to be detected in the image and the ground-truth is shown in Fig. 6(c).

TABLE I

AUC (%) VALUES OF THE SSBSTD WITH DIFFERENT PARAMETERS N_b FOR FOUR HYPERSPECTRAL DATA SETS

N_b	AVIRIS	HYDICE	Cri	Viareggio
100	98.6926	93.2919	93.7951	99.8998
200	99.2555	98.3940	**95.4584**	99.9621
300	99.2972	98.3095	95.0529	**99.9735**
400	**99.3241**	**98.6281**	94.8586	99.9717
500	98.6313	98.1819	94.8463	99.9608
600	98.6273	97.7467	94.9132	99.9713
700	98.7850	95.7050	94.7297	99.9710
800	98.7537	95.8033	94.6077	99.9688
900	98.3012	94.0102	94.3314	99.9710
1000	98.3174	93.5919	94.1963	99.9719

B. Parameter Analysis for the SSBSTD

This section discusses the influence of parameters on the detection performance of SSBSTD detector for the four HSIs.

In SSBSTD, there are three crucial parameters.

The first crucial parameter is t because it decides the amount of target samples. In this article, we obtain optimal t for four different HSIs based on our previous work [42]. Good detection performances can be achieved under $t = 2.9$ for the AVIRIS data set, $t = 3.4$ for the HYDICE data set, $t = 0.28$ for the Cri data set, and $t = 1.1$ for the Viareggio data set in this article.

Another critical parameter in SSBSTD is N_b that indicates the amount of background samples. In SSBSTD, we construct a global over-complete background dictionary based on classification; background training samples are those typical pixels selected from each cluster. Enough typical pixels in each cluster should be taken to make sure each pixel in HSI can obtain an appropriate representation and lead us to good detection performance. In this article, the range N_b is roughly set as [100 200 300 400 500 600 700 800 900 1000] for four data sets to test the effect of the parameter N_b on the SSBSTD. Table I demonstrates the sensitivity of the proposed SSBSTD over a wide range of N_b.

In this article, a widely used receiver operating characteristic (ROC) [52] curve is severed as a quantitative evaluation tool for assessing the detection performance. ROC curve can provide a threshold-free detection performance comparison by detecting a continuous curve of probability or false alarm rate [53]. Under the same false alarm level, algorithms with higher detection rates have better detection performance. The area enclosed by the ROC curve and coordinate axis which labeled as area under curve (AUC) value is also employed to assess the detection ability. Algorithms with larger AUC demonstrate better detection ability. Table I presents the AUC with different parameters N_b on four hyperspectral data for the SSBSTD.

In general, the AUC improves as N_b increases, but when N_b is too large, the AUC value goes down which means a worse detection performance. What is the reason? For a small N_b, there are few typical pixels in the background dictionary, so not all of the background pixels are well represented by these few typical pixels. As N_b increases, the background dictionary contains enough typical pixels of each feature, then each background pixel can obtain a good representation

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TABLE II
AUC (%) PERFORMANCES WITH DIFFERENT PARAMETERS Kb AS WELL AS Kt FOR AVIRIS DATA SET

Kb	Kt									
	2	4	6	8	10	12	14	16	18	20
10	48.9518	85.5029	95.6674	98.0279	98.2509	97.5778	96.9886	96.6328	96.4644	96.3438
12	49.5066	86.4916	96.2191	98.5166	99.1459	98.8361	98.5051	98.209	98.0064	97.8637
14	49.9310	87.0871	96.5216	98.6374	99.2996	99.1861	99.0264	98.8521	98.6948	98.5963
16	50.2184	87.5001	96.6734	98.6734	99.3352	99.2628	99.1717	99.0812	99.0048	98.9283
18	50.4325	87.7992	96.7744	98.6860	**99.3353**	99.2564	99.2184	99.1641	99.1144	99.0686
20	50.5809	88.0420	96.8291	98.6876	99.3241	99.2523	99.1984	99.1605	99.1417	99.1172
22	50.6805	88.2281	96.8820	98.6938	99.3091	99.2543	99.1989	99.1540	99.1257	99.1052
24	50.7066	88.2578	96.8039	98.6028	99.2081	99.1506	99.0995	99.0562	99.0345	99.0189
26	50.7250	88.3004	96.7623	98.5457	99.1395	99.0814	99.0306	98.9907	98.9709	98.9515
28	50.7016	88.2572	96.6554	98.4220	99.0122	98.9536	98.9022	98.8635	98.8453	98.8289
30	50.6614	88.1836	96.5287	98.2860	98.8715	98.8118	98.7629	98.7234	98.7044	98.6915

TABLE III
AUC (%) PERFORMANCES WITH DIFFERENT PARAMETERS Kb AS WELL AS Kt FOR VIAREGGIO DATA SET

Kb	Kt									
	2	4	6	8	10	12	14	16	18	20
10	99.6340	99.9742	99.9683	99.9679	99.9669	99.9665	99.9665	99.9665	99.9665	99.9665
12	99.6521	99.9769	99.9702	99.9702	99.9692	99.9690	99.9690	99.9690	99.9690	99.9690
14	99.6660	99.9790	99.9738	99.9721	99.9713	99.9713	99.9713	99.9713	99.9713	99.9713
16	99.6719	99.9800	99.9746	99.9731	99.9727	99.9725	99.9725	99.9725	99.9725	99.9725
18	99.6775	99.9802	99.9742	99.9727	99.9731	99.9727	99.9727	99.9727	99.9727	99.9727
20	99.6794	**99.9806**	99.9742	99.9733	99.9735	99.9735	99.9735	99.9735	99.9735	99.9735
22	99.6779	99.9779	99.9715	99.9715	99.9713	99.9713	99.9713	99.9713	99.9713	99.9713
24	99.6763	99.9725	99.9656	99.9660	99.9656	99.9656	99.9656	99.9656	99.9656	99.9656
26	99.4813	99.7923	99.7863	99.7863	99.7858	99.7858	99.7858	99.7858	99.7858	99.7858
28	99.3258	99.6383	99.6327	99.6325	99.6321	99.6321	99.6321	99.6321	99.6321	99.6321
30	99.2242	99.5510	99.5454	99.5452	99.5452	99.5452	99.5452	99.5452	99.5452	99.5452

TABLE IV
AUC (%) PERFORMANCES WITH DIFFERENT PARAMETERS Kb AS WELL AS Kt FOR HYDICE DATA SET

Kb	Kt									
	2	4	6	8	10	12	14	16	18	20
30	69.6612	90.4312	97.0133	98.5002	98.8005	98.8040	98.7926	98.7690	98.7593	98.7562
32	69.6683	90.4881	97.0352	98.5183	98.8107	98.8129	98.8062	98.7810	98.7710	98.7683
34	69.6750	90.5274	97.0536	98.5288	98.8174	98.8212	98.8112	98.7864	98.7755	98.7714
36	69.6774	90.5500	97.0645	98.5364	98.8200	98.8264	98.8145	98.7926	98.7807	98.7771
38	69.6793	90.5636	97.0721	98.5455	98.8233	98.8293	98.8176	98.7955	98.7850	98.7805
40	69.6812	90.5748	97.0752	98.5462	98.8248	98.8314	98.8193	98.7964	98.7855	98.7819
42	69.6817	90.5805	97.0767	98.5474	98.8257	98.8324	98.8200	98.7993	98.7869	98.7833
44	69.6821	90.5831	97.0779	98.5481	98.8260	98.8326	98.8205	98.8000	98.7881	98.7840
46	69.6821	90.5862	97.0783	98.5481	98.8260	98.8333	98.8205	98.8000	98.7881	98.7840
48	69.6821	90.5871	97.0783	98.5481	98.8260	**98.8336**	98.8205	98.8000	98.7886	98.7840
50	69.6821	90.5871	97.0783	98.5481	98.8260	98.8336	98.8205	98.8000	98.7886	98.7840

and finally a good detection performance can be achieved. For a large N_b, there may be some pixels contaminated by noise in it, causing a bad representation for some background pixels, which may reduce the discriminative power and achieve a worse detection performance. According to Table I, the optimal values of background training samples N_b can be obtained for four hyperspectral data sets. They are $N_b = 400$ for AVIRIS and HYDICE data set, $N_b = 200$ for the Cri data set, and $N_b = 300$ for the Viareggio data set. These optimal values N_b are also used in SSBSTD for final detection.

The third critical parameter is the sparsity level. In our proposed BSR, there are two sparsity levels under the two hypotheses (Kb for H_0 and Kt for H_1). In other binary hypotheses-based sparse representation models such as SRBBH, both hypotheses employ the same sparsity constraint which means Kb should be equal to Kt. However, in our proposed detector, since we use a global over-complete background dictionary to sparsely represent background pixels instead of a local one, higher sparsity level should be used to make sure an adequate fitting for each background pixel.

Herein, the validity of the constraint on the sparsity level is verified via the detection performance comparison of two hypotheses with different sparsity levels. Due to the fact that we have no specific approach to decide the optimum value for sparsity level, we tune the parameter experientially and manually by setting the range of Kt as [2 4 6 8 10 12 14 16 18 20] for all four hyperspectral data sets, the range of Kb as [10 12 14 16 18 20 22 24 26 28 30] for AVIRIS and Viareggio data set, and the range of Kb as [30 32 34 36 38 40 42 44 46 48 50] for HYDICE and Cri data set. AUC performances of proposed SSBSTD with different parameters Kb as well as different Kt for all HSI data are

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TABLE V
AUC (%) PERFORMANCES WITH DIFFERENT PARAMETERS Kb AS WELL AS Kt FOR CRI DATA SET

Kb	Kt									
	2	4	6	8	10	12	14	16	18	20
30	93.1211	95.0649	95.4364	95.5135	95.5381	95.5027	95.4315	95.2274	93.5714	92.5196
32	93.1299	95.0722	95.4441	95.5216	95.5464	95.5112	95.4400	95.2358	93.5784	92.5258
34	93.1373	95.0784	95.4505	95.5281	95.5530	95.5179	95.4468	95.2423	93.5838	92.5304
36	93.1447	95.0848	95.4571	95.5348	95.5597	95.5248	95.4537	95.2491	93.5895	92.5356
38	93.1528	95.0919	95.4643	95.5421	95.5669	95.5324	95.4615	95.2567	93.5956	92.5410
40	93.1629	95.1003	95.4727	95.5504	95.5755	95.5410	95.4702	95.2653	93.6029	92.5475
42	93.1773	95.1119	95.4842	95.5618	95.5868	95.5526	95.4820	95.2771	93.6126	92.5563
44	93.2000	95.1301	95.5020	95.5795	95.6042	95.5703	95.5001	95.2948	93.6274	92.5696
46	93.2268	95.1487	95.5174	95.5931	**95.6172**	95.5838	95.5140	95.3082	93.6363	92.5767
48	93.2195	95.1414	95.5101	95.5857	95.6098	95.5764	95.5066	95.3008	93.6291	92.5696
50	93.2195	95.1414	95.5101	95.5857	95.6098	95.5764	95.5066	95.3008	93.6291	92.5696

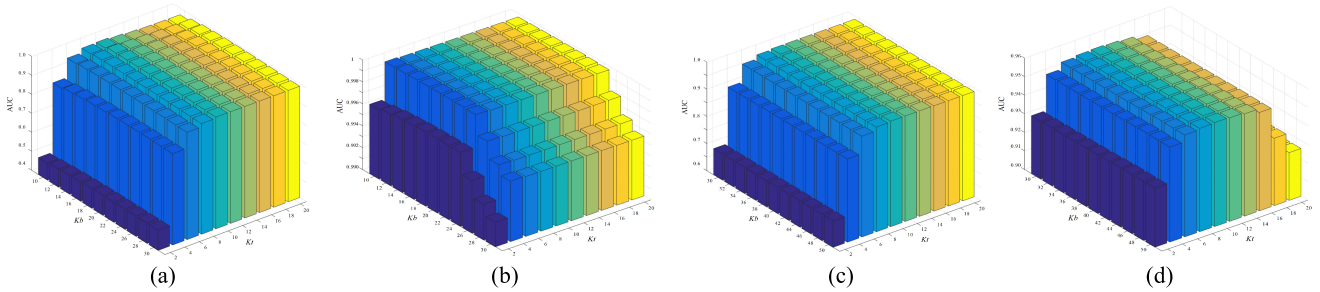

Fig. 7. Analysis of the sparsity level Kb and the sparsity level Kt on all data sets. (a) AVIRIS. (b) Viareggio. (c) HYDICE. (d) Cri.

provided in Tables II–V. According to Tables II–V, AUC with the two sparsity levels show the same pattern. In general, both for Kt and Kb, AUC value becomes larger when the sparsity level improves to a certain level, then it goes down when the sparsity level is too large. The reason can be expressed as follows. For a small sparsity level, the sparsity-based technique is simply reduced to template matching and it is more likely to cause the problem of under-fitting causing weakened detection performance. When the sparsity level is too large, the solution tends to be dense and also leads to weakened discrimination.

We also visualize the analysis of the sparsity level Kb and the sparsity level Kt on all data sets for providing a more intuitive perspective, as is shown in Fig. 7(a)–(d). According to Tables II–V and Fig. 7(a)–(d), the optimal values of two sparsity levels to proposed SSBSTD are $Kt = 10$ and $Kb = 18$ for AVIRIS data set, $Kt = 12$ and $Kb = 48$ for HYDICE data set, $Kt = 10$ and $Kb = 46$ for Cri data set, and $Kt = 4$ and $Kb = 20$ for Viareggio data set. Through these optimal values of sparsity levels, we may find that Kb is always larger than Kt. That is the case that higher sparsity level should be used to make sure an adequate fitting for each background pixel because we use a global background dictionary instead of a local one in SSBSTD.

C. Detection Performance and Analysis

First of all, the effectiveness analysis of each portion of our proposed SSBSTD is discussed. Our SSBSTD consists of three main portions: a new method for TDC, a novel approach based on classification to generate a global over-complete background dictionary, and the final classifier BSR model.

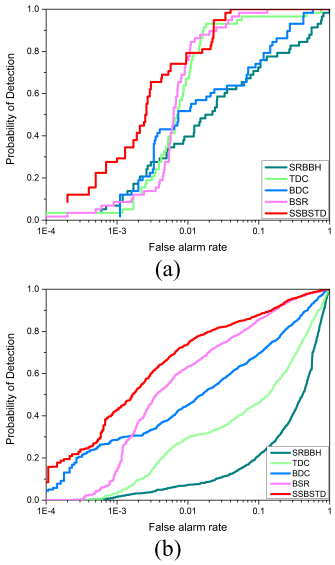

Fig. 8. ROC curves of effectiveness analysis in SSBSTD for two data sets. (a) AVIRIS. (b) Cri.

Two data sets AVIRIS and Cri are used to evaluate the effectiveness of each portion. We use SRBBH as the initial detector since SRBBH is a binary hypothesis-based model similar to our SSBSTD. SRBBH with the new method for TDC is used to test the efficiency of TDC in SSBSTD. The approach for BDC imposed on SRBBH serves as the BDC detector to estimate the efficiency of BDC. The last portion BSR model is also regarded as a detector to evaluate the effectiveness. Fig. 8 illustrates the ROC performance of effectiveness analysis in SSBSTD for two data sets. It is

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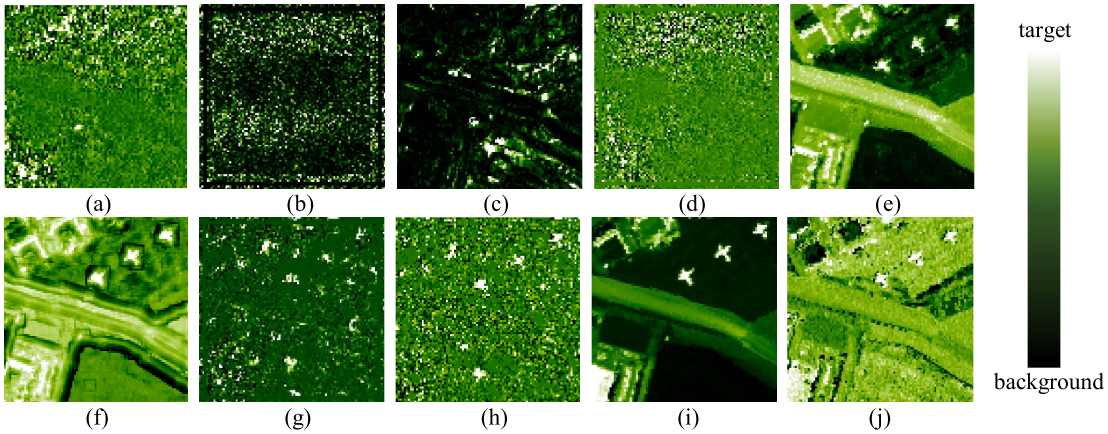

Fig. 9. 2-D color map of detection results for AVIRIS data set. (a) CEM, (b) ACE, (c) MSD, (d) SMF, (e) STD, (f) SASTD, (g) SRBBH, (h) HSSD, (i) CSCR, and (j) SSBSTD.

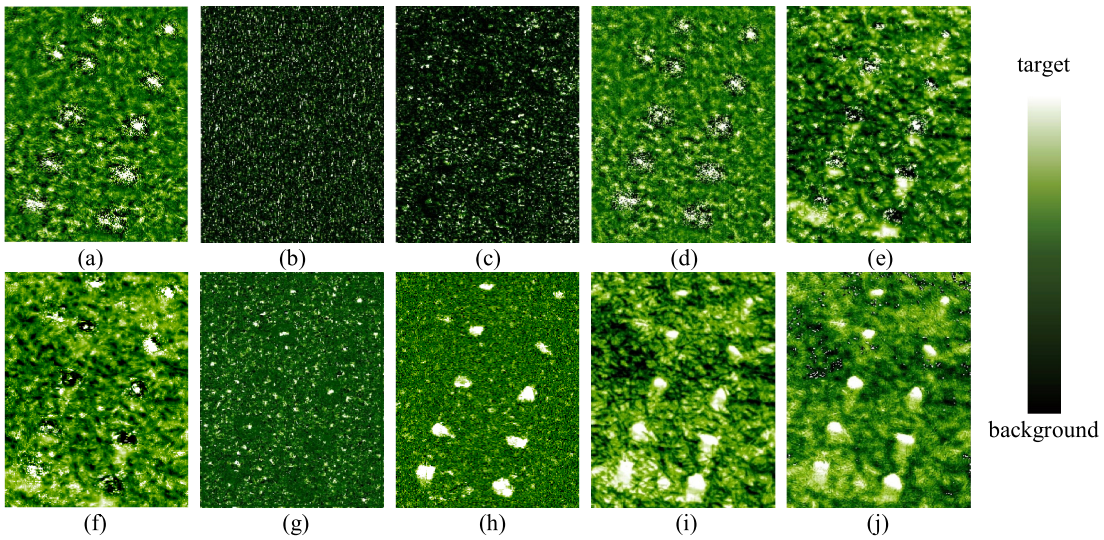

Fig. 10. 2-D color map of detection results for Cri data set. (a) CEM, (b) ACE, (c) MSD, (d) SMF, (e) STD, (f) SASTD, (g) SRBBH, (h) HSSD, (i) CSCR, and (j) SSBSTD.

apparent that our SSBSTD achieves the best performance since it demonstrates its quantitative superiority over the other four detectors because of the highest ROC curve. Compared with the SRBBH, both TDC and BDC yield a better detection result.

The reasons are as follows. For TDC, it can obtain enough target training samples for the TDC, thus leading a good representation of target pixels and finally enhancing the detection efficiency. For BDC, the global over-complete background dictionary is more complete which can guarantee each background pixel in the image scene obtains a better representation, then leads to better detection performance. The BSR model also demonstrates superior performance. The main reason is BSR only use background training samples to represent background pixels and target training samples to target pixels. It avoids the problem that the two sparse vectors affect each other, and get the adequate discriminative capacity to separate the target and the background. As a result, taking advantage of these three portions, our SSBSTD can separate targets from backgrounds and effectively detect targets, thereby achieving an outstanding detection result.

The proposed SSBSTD is evaluated and contrasted to the subjacent advanced algorithms: 1) the local CEM; 2) the local ACE; 3) the local MSD; 4) the local SMFs; 5) the basic STD; 6) the SASTD; 7) the SRBBH detector; 8) the HSSD; and 9) the combined sparse and collaborative representation (CSCR) detector. As for the detectors STD, SASTD, SRBBH, HSSD, and CSCR, the target pixels selected from the HSI are used as prior information to form the target dictionary, and for the other detectors, we use the same target spectrum as prior information for detection. In the local CEM, local ACE, local MSD, and local SMF, pixels in the dual window are utilized for the background covariance matrix. In STD, SASTD, SRBBH, HSSD, and CSCR, the pixels that are distributed in the dual window are used for the BDC.

First of all, the detection result maps of the SSBSTD for AVIRIS and Cri data set are presented in Figs. 9 and 10. It is obvious in Fig. 9 that for the AVIRIS data set, four classic detectors CEM, ACE, MSD, and SMF have bad visualized detection results, they cannot highlight most of the targets.

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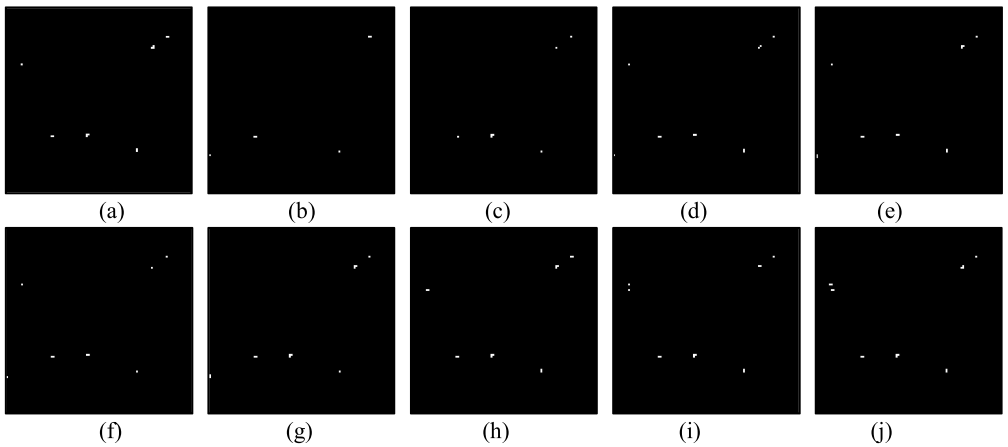

Fig. 11. Binary map of detection results for HYDICE data set. (a) CEM, (b) ACE, (c) MSD, (d) SMF, (e) STD, (f) SASTD, (g) SRBBH, (h) HSSD, (i) CSCR, and (j) SSBSTD.

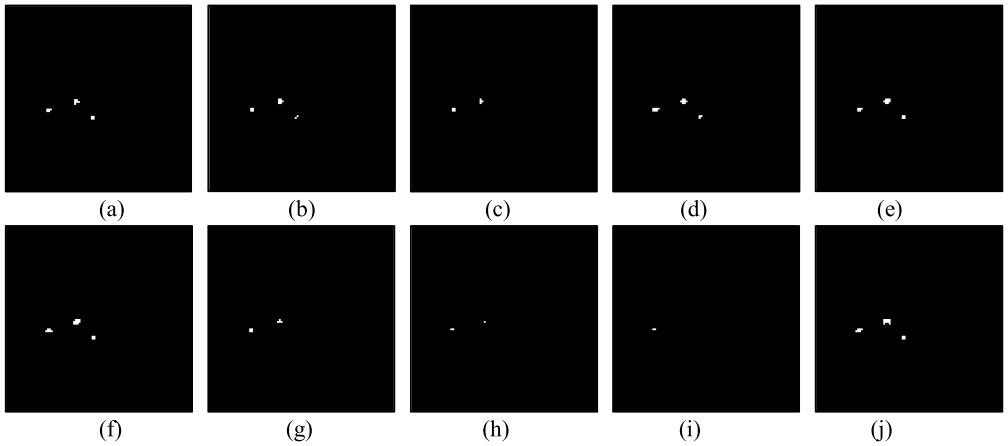

Fig. 12. Binary map of detection results for Viareggio data set. (a) CEM, (b) ACE, (c) MSD, (d) SMF, (e) STD, (f) SASTD, (g) SRBBH, (h) HSSD, (i) CSCR, and (j) SSBSTD.

The other detectors all achieve a nice performance in highlighting targets, especially the detectors CSCR and SSBSTD, but SSBSTD's performance is even more outstanding since CSCR cannot suppress the background pixels, that is the buildings in the bottom-left corner. For the Cri data set, it is obvious that only HSSD, CSCR, and our proposed SSBSTD are able to detect all targets in high confidence, others can detect only a few of them. Among these three detectors, HSSD and CSCR fail to subdue the background pixels around the targets, and SSBSTD can accurately suppress them. Since the 2-D color map is not appropriate to show the details of detection results for HYDICE and Viareggio data sets, detection results in binary map for the two data sets are provided in Figs. 11 and 12. In the binary maps, the real target pixels can be highlighted as light pixels only when their corresponding output values of the detection results are the biggest ones in the front. For the HYDICE data set, many detectors can detect most of the targets except ACE and MSD detector. Although these detectors perform well in detecting most of the targets, they fail to detect targets in the top-left corner, especially the SRBBH detector. Our SSBSTD can correctly detect them. Therefore, the proposed SSBSTD achieves a better detection performance among these detectors in the HYDICE data set. Fig. 12 illustrates the results of detection in binary maps for the Viareggio data set. It is evident that many detectors have a good result in detecting targets, such as ACE, SMF, STD, SASTD, and SSBSTD. HSSD and CSCR detectors have poor performance since they can only detect few targets. Compared with other nine detectors, our proposed SSBSTD can correctly highlight as many target pixels as possible, which means a promising detection result. In general, the proposed SSBSTD method can accurately detect the targets. SSBSTD is benefited from adequate target samples and a global BDC. Adequate target training samples make sure that each target pixel can get a fairly good representation and a global over-complete background dictionary contributes to background pixels' representation. Therefore, they finally enhance the detection performances.

In addition, for the purpose of quantitatively comparing the detection performance of SSBSTD algorithm with aforesaid nine detectors, the ROC curves for four data are provided in Fig. 13(a)–(d) and the AUC of all the detectors

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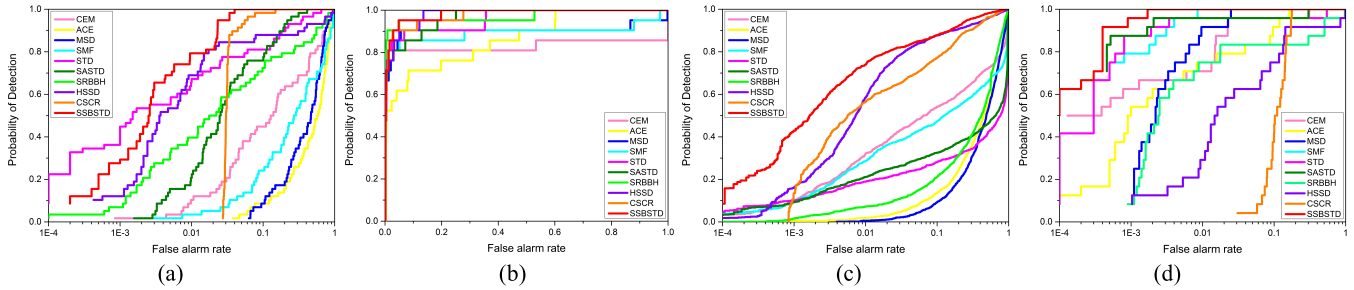

Fig. 13. ROC performance of all detectors for four data sets. (a) AVIRIS, (b) HYDICE, (c) Cri, and (d) Viareggio.

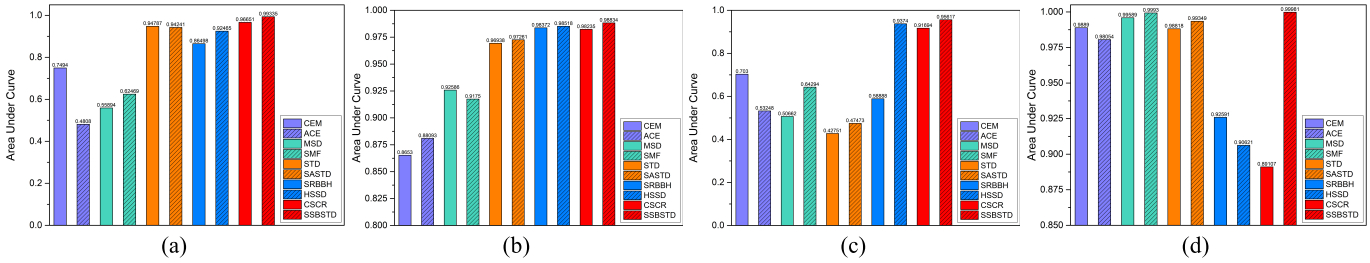

Fig. 14. AUC performance of all detectors for four data sets. (a) AVIRIS, (b) HYDICE, (c) Cri, and (d) Viareggio.

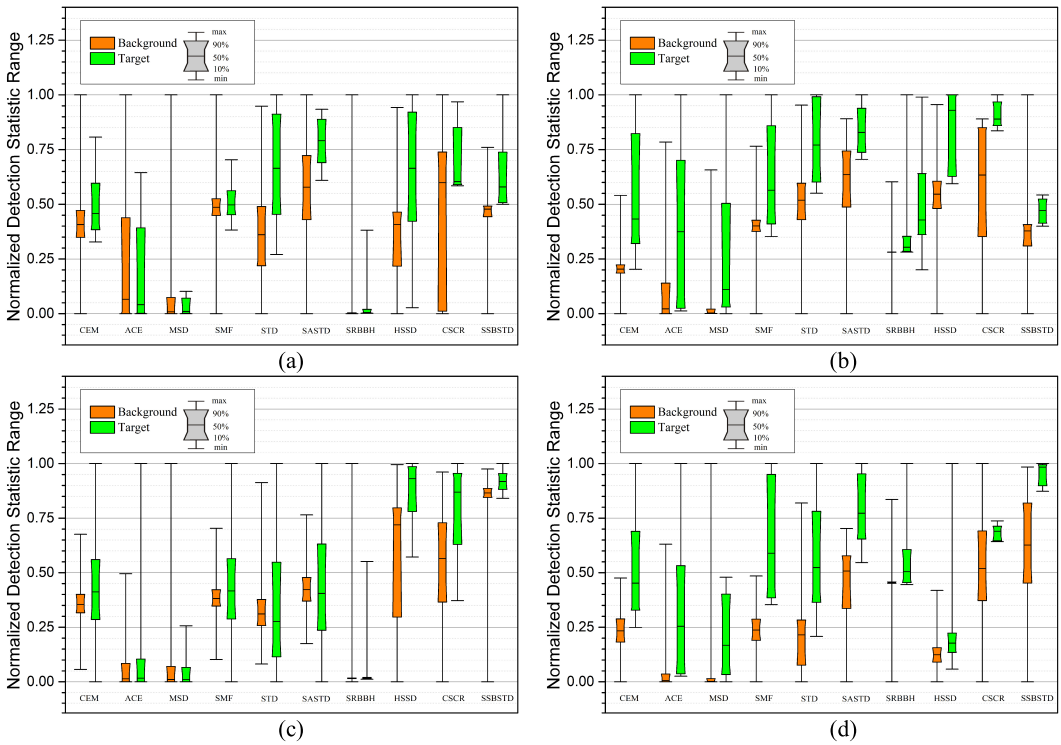

Fig. 15. Separability maps of all detectors for four data sets. (a) AVIRIS, (b) HYDICE, (c) Cri, and (d) Viareggio.

are presented in Fig. 14(a)–(d). ROC curves and AUC values are presented in Figs. 13(a) and 14(a) for the AVIRIS data. From Fig. 13(a), we can see that four classic detectors CEM, ACE, MSD, and SMF perform poorly, as they fail to correctly highlight the targets, generating high false alarms. In the sparsity-based detectors, the ROC curves of SASTD, SRBBH, HSSD, and CSCR are under SSBSTD's. Although the ROC curve of STD is ahead of SSBSTD's at the beginning, it falls a long way behind later. Besides, it is obvious that SSBSTD' AUC value is higher than STD in Fig. 14(a). Therefore, SSBSTD obtains the outstanding performance in target detection in the AVIRIS data set. Figs. 13(b) and 14(b) show the ROC curves and AUC values for the HYDICE data set. From Fig. 13(b), it seems that all the sparsity-based detectors except SASTD and HSSD have relatively good detection results, but their ROC curves are intertwined, so we cannot tell which one performs better just from the ROC curve. However, the AUC value of SSBSTD is the maximum in Fig. 14(b),

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so SSBSTD also reaches the most outstanding performance among them in the HYDICE data set. For the Cri data, we can see evidently that the SSBSTD is superior to all the other detectors since the ROC curve always takes the leading position in Fig. 13(c) and the AUC value is much higher than others. For the Viareggio data set, we can see from Fig. 13(d) that our proposed SSBSTD works effectively. Contrast to the other detectors, it can rationally keep the false alarm rate and detects targets at the same time as its ROC curve lies in the top-left corner. Fig. 14(d) gives the AUC values of detection results for the Viareggio data. It is evident in Fig. 14(d) that the AUC values of SMF and SSBSTD are very close to 1, which demonstrates almost perfect performance. SSBSTD achieves the highest detection accuracy since the AUC value of it is the maximum among these detectors.

Moreover, for the purpose of testing the validity of SSBSTD in separating targets and backgrounds, we provide separability maps that are evaluated by the normalized detection statistic range, as shown in Fig. 15(a)–(d). On the basis of the ground-truth information, pixels with output value are divided into targets and backgrounds, and all the detection result values are normalized to [0, 1] in two classes, respectively. A box chart is used to represent separability between targets and backgrounds. In the box, the orange one presents the range of background class, while the green one expresses the target class. Boxes illustrate the main portions of the pixels between 10% and 90% in their class. Better performance in separability means a larger separability distance between target and background class and a smaller background distribution range. For the AVIRIS and Cri data sets, it is evident that all the detectors except SSBSTD fail to highlight targets from backgrounds, as the green boxes and orange boxes are overlapped, and only our proposed SSBSTD has a certain separability between two boxes. Therefore, the SSBSTD detector performs well on separating targets from backgrounds in these two data sets. For the HYDICE data set, ACE, SMF, and SRBBH perform poorly as there exists no gap between two boxes, and the other methods all have a gap especially CEM and HSSD. The performance of our proposed SSBSTD is third only to CEM and HSSD since it has a certain separability between two boxes as well as a small background distribution range. In the Viareggio data set, we can see that most of the detectors have a gap between two boxes, especially SMF, STD, SASTD, and SSBSTD. However, SMF, STD, and SASTD cannot effectively separate the two classes in AVIRIS and Cri data sets due to the fact that there are obvious overlaps between two classes in Fig. 15(a) and (c). In contrast, SSBSTD shows an obvious gap between the two boxes, which means a promising result on highlighting targets from backgrounds.

IV. Conclusion

In this article, the SSBSTD detector is proposed as an efficient algorithm for hyperspectral target detection. The SSBSTD relies on the BSR model to generate discriminative ability, which can effectively separate targets from backgrounds. For the sake of obtaining sufficient target samples in the sparse representation model, a new method for constructing the target dictionary via only a given target spectrum is introduced. Meanwhile, by utilizing a novel approach based on classification, SSBSTD gets a global over-complete background dictionary which is more complete and can make sure each background pixel in the image scene obtains a better representation. Extensive experiments with four benchmark HSIs prove the performance of the SSBSTD algorithm.

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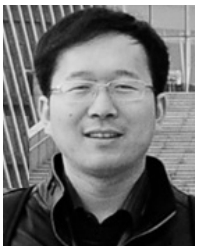

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